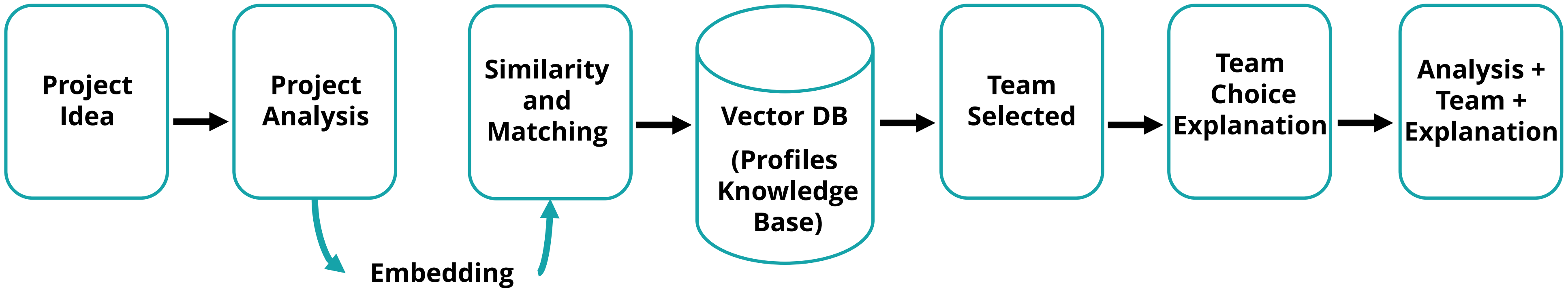
Final Project on Smart Applications in the Master of Software Engineering

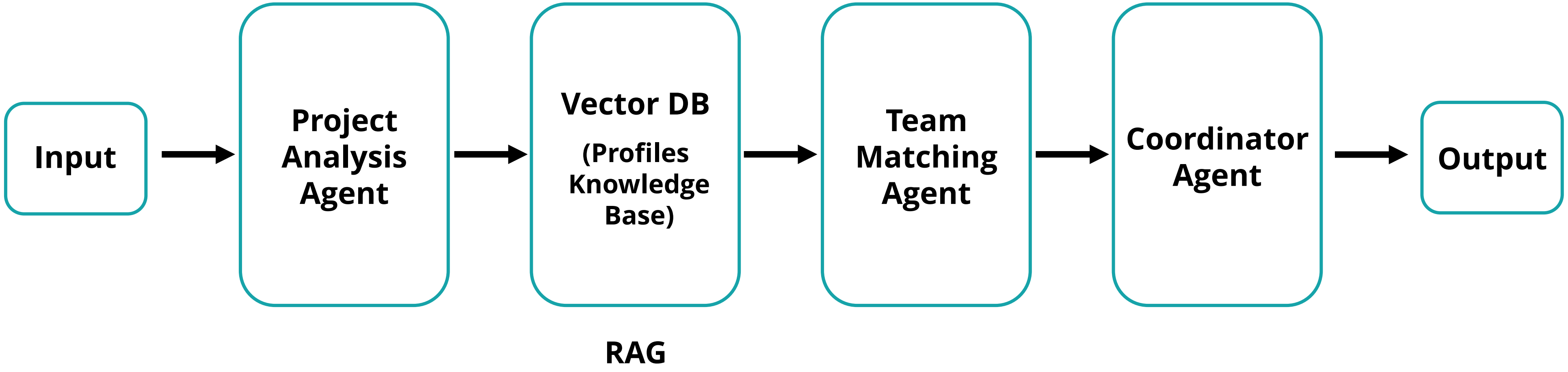
Selecting Software Development Team using AI

Supervised by Dr. Wesam Alnabki

Prepared by E. Salam Alkhawwam & E. Amal Al-Khalaf


Project Idea
Project Analysis
Similarity and Matching
Vector DB
(Profiles Knowledge Base)
Team Selected
Team Choice Explanation
Analysis + Team + Explanation
Embedding

Semi-Agentic Architecture



Python + Fast API + HTML + CSS

Python :

- Relies on AI / LLMs / RAG
- Rapid Development
- Seamless Integration with AI Tools (LangChain, Hugging Face, and vector databases)

Fast API :

- High Performance
- Ideal for I/O-bound AI workloads and external model calls.
- Strong Data Validation (Type-safe request and response validation using Pydantic)
- Developer-Friendly (Minimal boilerplate with clear and readable code)

HTML+CSS :

- Lightweight & Fast & Easy to Use
- Cross-Browser Compatibility

LLM Orchestration Framework (LangChain)

- Flexible LLM Orchestration Framework
- Seamless RAG Integration (like Chroma)
- Multi-LLM Support
- Prompt Management & Experimentation
- Agent & Tool Integration
- Asynchronous & Scalable
- Open-Source & Extensible

Local LLM Inference Engine (Ollama)

- Open-Source & Free
- Fast Development Workflow
- Cross-Platform Support
- Seamless LangChain Integration & RAG pipelines.
- Can be used as a drop-in replacement for OpenAI APIs
- Simple Local LLM Runtime (Run LLM locally with a single command)
- Zero-Configuration Setup (No Docker, cloud accounts, or complex infrastructure required)
- Local-First & Privacy-Friendly (All data and inference stay on the local machine)
- Built-in Model Management (Easy download, versioning, and switching between models)

Profile Knowledge Base

Meta Data

- ID
- Name
- Role
- Seniority
- Years Experience
- Availability
- Time zone
- Languages
- Skills
- Frameworks
- Domains
- Tools
- Project Types
- Summary

Embedding

Query

Meta Data

- ID
- Name
- Role
- Seniority
- Years Experience
- Availability
- Time zone
- Languages
- Skills
- Frameworks
- Domains
- Tools
- Project Types
- Summary

Embedding

Json

- Lightweight and easy to use
- No database setup or maintenance required
- Human-readable and easy to debug
- Flexible schema suitable for evolving data
- Ideal for small and static datasets
- Fast prototyping and rapid development
- Easily integrated with AI pipelines and LLM workflows

Vector DB (Chroma DB)

- Lightweight & Easy to Use & Simple setup
- Local-First Design without requiring a separate server
- Excellent for Rapid Prototyping & RAG pipelines
- Native Integration with LLM Frameworks
- Seamless support for LangChain
- Persistent Storage
- Metadata Filtering Support
- Fast semantic search for small to medium-scale datasets
- Open-Source & Free
- Low Infrastructure Overhead (No need for Docker, Kubernetes, or cloud services)

Embedding Model (Nomic)

Advantages :
- Fast embedding generation
- Lightweight & local-friendly
- Easy integration with Python & LangChain
- Free & open-source
- Ideal for small-to-medium projects

Disadvantages :
- Lower accuracy than newer models
- Limited multilingual support
- Not suitable for very large datasets
- No built-in fine-tuning support

Embedding Model (BGE)

Advantages :

- High accuracy embeddings
- Excellent for semantic search
- Handles large datasets efficiently
- Multilingual support
- Compatible with modern LLM pipelines

Disadvantages :

- Requires more computational resources
- Slower than lightweight models
- Larger model size
- Overkill for small-scale projects

Nomic vs BGE

Evaluation (precision@k) - simple 1

```
role = "Backend Engineer"
seniority = "Senior"
skills = ["API design", "scalable systems", "health data"]
ground_truth_ids = ["backend_004", "backend_010", "backend_001"]
```

```
📊 Results:
   model  precision@k  recall@k  MRR                                   predicted_ids
0  Nomic     0.000000  0.000000  0.0  [frontend_008, fullstack_004, backend_003]
1    BGE     0.666667  0.666667  1.0     [backend_001, backend_010, backend_008]
```

Bge-m3

Nomic vs BGE

Evaluation (precision@k) - simple 2

```
role = "Frontend Engineer"
seniority = "Mid"
skills = ["dashboard UI", "accessibility"]
ground_truth_ids = ["frontend_009", "frontend_005", "frontend_008"]
```

```
📊 Results:
   model  precision@k  recall@k       MRR                                   predicted_ids
0  Nomic     0.000000  0.000000  0.000000  [frontend_003, frontend_002, frontend_004]
1    BGE     0.333333  0.333333  0.333333  [frontend_003, frontend_002, frontend_009]
```

↑ Bge-m3

Nomic vs BGE

Evaluation (precision@k) - simple 3

```
role = "Fullstack Developer"
seniority = "Senior"
skills = ["web apps", "scalability"]
ground_truth_ids = ["fullstack_001", "fullstack_010", "fullstack_006"]
```

```
📊 Results:
   model  precision@k  recall@k  MRR                                  predicted_ids
0  Nomic     0.333333  0.333333  0.5    [frontend_007, fullstack_001, frontend_006]
1    BGE     1.000000  1.000000  1.0  [fullstack_010, fullstack_006, fullstack_001]
```

↑ Bge-m3

Nomic vs BGE

Evaluation (precision@k) - simple 4

```
role = "Machine Learning Engineer"
seniority = "Senior"
skills = ["fraud detection", "NLP"]
ground_truth_ids = ["ai_001", "ai_010", "ai_004"]
```

```
📊 Results:
   model  precision@k  recall@k  MRR                                   predicted_ids
0  Nomic     0.000000  0.000000  0.0  [security_010, security_003, security_002]
1    BGE     0.666667  0.666667  1.0                    [ai_001, ai_010, ai_006]
```

Bge-m3

Nomic vs BGE

Evaluation (precision@k) - simple 5

```
role = "Security Engineer"
seniority = "Senior"
skills = ["application security", "fintech compliance"]
ground_truth_ids = ["security_001", "security_006", "security_008"]
```

```
📊 Results:
   model  precision@k  recall@k  MRR                                  predicted_ids
0  Nomic     0.666667  0.666667  0.5  [security_003, security_008, security_001]
1    BGE     0.666667  0.666667  1.0  [security_006, security_001, security_010]
```

↑ Nomic + Bge-m3

Nomic vs BGE

Evaluation (precision@k) - simple 6

```
role = "DevOps Engineer"
seniority = "Senior"
skills = ["secure deployments", "monitoring"]
ground_truth_ids = ["devops_004", "devops_008", "devops_006"]
```

```
📊 Results:
   model  precision@k  recall@k       MRR                            predicted_ids
0  Nomic     0.333333  0.333333  0.333333  [security_004, security_010, devops_004]
1    BGE     0.333333  0.333333  0.500000      [devops_010, devops_006, devops_001]
```

↑ Nomic + Bge-m3

Nomic vs BGE

Evaluation (precision@k) - simple 7

```
role = "NLP Engineer"
seniority = "Mid"
skills = ["information retrieval", "text summarization"]
ground_truth_ids = ["ai_007", "ai_003", "ai_005"]
```

```
📊 Results:
   model  precision@k  recall@k  MRR                    predicted_ids
0  Nomic     0.333333  0.333333  1.0  [ai_007, data_003, data_007]
1    BGE     0.666667  0.666667  1.0    [ai_007, data_007, ai_005]
```

↑ Bge-m3

Nomic vs BGE

Evaluation (precision@k) - simple 8

```
role = "Data Engineer"
seniority = "Mid"
skills = ["data ingestion", "pipelines"]
ground_truth_ids = ["data_010", "data_005", "data_002"]
```

```
📊 Results:
   model  precision@k  recall@k  MRR                     predicted_ids
0  Nomic     0.666667  0.666667  1.0  [data_002, data_007, data_010]
1    BGE     0.666667  0.666667  0.5  [data_007, data_005, data_002]
```

Nomic + Bge-m3

Nomic vs BGE

Evaluation (precision@k) - simple 9

```
role = "DevOps Engineer"
seniority = "Senior"
skills = ["cloud", "security", "kubernetes"]
ground_truth_ids = ["devops_001", "devops_006", "devops_010"]
```

```
📊 Results:
   model  precision@k  recall@k  MRR                                predicted_ids
0  Nomic     0.000000  0.000000  0.0  [security_002, security_009, security_005]
1    BGE     0.666667  0.666667  1.0        [devops_010, devops_006, devops_008]
```

↑ Bge-m3

Nomic vs BGE

Evaluation (precision@k) - simple 10

```
role = "Frontend Engineer"
seniority = "Senior"
skills = ["complex dashboards", "performance"]
ground_truth_ids = ["frontend_007", "frontend_001", "frontend_004"]
```

```
📊 Results:
   model  precision@k  recall@k  MRR                                predicted_ids
0  Nomic     0.333333  0.333333  1.0     [frontend_007, frontend_009, data_006]
1    BGE     0.333333  0.333333  0.5  [backend_001, frontend_007, frontend_009]
```

Nomic + Bge-m3

Nomic vs BGE

Evaluation (precision@k) - 10 simples :

precision@k (10 simples) – nomic : 0.24
precision@k (10 simples) – bge-m3 : 0.55

↑ **Bge-m3**

BGE model is more accurate than Nomic model

LLM (llama3-7B)

Advantages :

- Open-source and locally usable
- Lightweight relative to larger LLaMA models
- Excellent general language understanding
- Compatible with RAG frameworks (LangChain, ChromaDB)

Disadvantages :

- Lower accuracy than larger models in complex tasks
- Requires moderate GPU resources
- Limited multilingual performance without further training

LLM (Mistral)

Advantages :

- Excellent semantic understanding & text generation
- Faster than comparable LLaMA models
- Lightweight and local-friendly with quantization
- Open-source and integrates well with RAG pipelines
- Good multitask performance across general tasks

Disadvantages :

- Less fine-tuning support than LLaMA ecosystem
- Requires modern GPU for smooth inference
- Responses can be less controlled in certain prompts

Llama3 LLM : 5 Min
Embedding Model : BGE

Team Builder using AI

```
I want to build a SaaS platform for managing online payments
for small businesses. The system needs to be scalable,
secure, and API-first.
```

Build Team

Team Builder using AI

📝 Project Analysis


```
{
  "project_type": "SaaS",
  "domains": [
    "online_payments"
  ],
  "required_roles": [
    {
      "role": "Software Architect",
      "seniority": "Senior",
      "skills": [
        "API design",
        "Security"
      ]
    },
    {
      "role": "DevOps Engineer",
      "seniority": "Mid-level",
      "skills": [
        "Containerization",
        "Cloud deployment"
      ]
    }
  ],
  "tech_stack": [
    "Java",
    "Spring Boot",
    "AWS"
  ],
  "team_size": 5
}
```


🤖 Team Members

Sofia Rossi (Software Architect)

Seniority: Mid

Reason: Strong API design skills and experience with Security

Jin Woo Kim (DevOps Engineer)

Seniority: Senior

Reason: Experience with Containerization and Cloud deployment, strong DevOps skills

📌 Coordinator Report

Based on the provided project analysis and selected team, I'll analyze each member's selection, highlight team strengths and weaknesses, identify potential issues, and provide actionable recommendations.

Why each member was selected:

Sofia Rossi (backend_005) - As a mid-level Software Architect, Sofia brings strong API design skills and experience with Security, which aligns well with the project requirements. Her seniority level is lower than expected, but her skills make up for it.

Jin Woo Kim (devops_010) - With his senior-level DevOps Engineer role, Jin brings extensive experience in Containerization and Cloud deployment, making him an excellent fit for the DevOps role. His strong DevOps skills will help ensure smooth operations.

Team strengths:

* Sofia's API design expertise will lead to a well-structured software architecture.
* Jin's DevOps skills will enable efficient cloud deployment and containerization.
* The team has a good balance of technical skills, with both software development (Sofia) and infrastructure management (Jin).

Weaknesses or risks:

* Sofia's mid-level seniority might lead to some communication challenges with more senior stakeholders.
* Jin's senior-level experience may not be fully utilized in the current project scope.

Missing roles or seniority imbalance:

* The team lacks a dedicated front-end developer and a junior or entry-level role to fill the 5-person team size. This could result in an uneven workload distribution and limited opportunities for growth.
* There is no clear leadership or management presence on the team, which may lead to coordination challenges.

Actionable recommendations:

1. Consider adding a junior or entry-level developer to balance out the workload and provide opportunities for growth.
2. Identify potential candidates for leadership or management roles within the existing team members or consider hiring externally.
3. Provide additional training or mentorship for Sofia to help her grow into a more senior role, potentially leading to improved communication with stakeholders.

By addressing these areas, the team can better align with project requirements and optimize its performance.

Mistral LLM : 8 Min
Embedding Model : BGE

Team Builder using AI

```
I want to build a SaaS platform for managing online payments
for small businesses. The system needs to be scalable,
secure, and API-first.
```

Build Team

Team Builder using AI

📝 Project Analysis


```
{
  "project_type": "SaaS Platform",
  "domains": [
    "Online Payments",
    "Small Businesses"
  ],
  "required_roles": [
    {
      "role": "Software Architect",
      "seniority": "Senior",
      "skills": [
        "Scalability",
        "Security",
        "API Design"
      ]
    },
    {
      "role": "Full Stack Developer",
      "seniority": "Mid-Level",
      "skills": [
        "JavaScript",
        "Node.js",
        "RESTful APIs"
      ]
    }
  ],
  "tech_stack": [
    "Node.js",
    "Express.js",
    "PostgreSQL",
    "Stripe API"
  ],
  "team_size": 5
}
```

Team Members

Jamal Thompson (Software Architect)

Seniority: Senior

Reason: Strong background in backend development and extensive experience, making him a suitable candidate for the Software Architect role.

Rajesh Iyer (Full Stack Developer)

Seniority: Senior

Reason: Experienced Full Stack Developer with a strong background in JavaScript, Node.js, and RESTful APIs, aligning well with the project's requirements.

Nadia Petrovič (Full Stack Developer)

Seniority: Senior

Reason: Senior Full Stack Developer with 10 years of experience, capable of contributing to the project's scalability and security requirements.

📌 Coordinator Report

1. Jamal Thompson was selected for the Software Architect role due to his strong background in backend development and extensive experience, making him a suitable candidate for designing scalable, secure, and robust solutions that meet the project's requirements.

Rajesh Iyer was chosen as the Full Stack Developer due to his experienced proficiency in JavaScript, Node.js, and RESTful APIs, ensuring he can effectively contribute to the development of the SaaS platform. Nadia Petrovič was also selected as a Full Stack Developer, having 10 years of experience that will enable her to contribute significantly to the project's scalability and security requirements.

2. The team brings together a strong combination of senior-level software architects and full stack developers with diverse skill sets in backend development, API design, and security. This synergy should foster efficient collaboration, high-quality code production, and effective problem-solving.

3. Potential weaknesses or risks include the team's lack of experience in specific areas such as frontend development and user interface/user experience (UI/UX) design. This could result in a suboptimal user experience for the small businesses using the SaaS platform. Additionally, there is no dedicated project manager to oversee the timeline, budget, and resources, which might lead to delays or mismanagement.

4. Although the selected team covers the required roles of Software Architect and Full Stack Developer, there is currently no identified team member with expertise in frontend development, which may require additional hiring or assigning tasks to external contractors. The current team composition also lacks a balance between junior and senior-level developers, potentially limiting opportunities for knowledge sharing and mentoring within the team.

5. To address these challenges, consider the following actionable recommendations:
   - Hire or assign a frontend developer with expertise in modern UI/UX design to enhance the user experience of the SaaS platform.
   - Appoint a dedicated project manager to oversee the project's timeline, budget, and resources, ensuring efficient execution and minimizing potential delays.
   - Encourage knowledge sharing and mentoring within the team by fostering an environment that promotes collaboration and learning among junior and senior-level developers.

Llama3 LLM : 6 Min
Embedding Model : Nomic

Team Builder using AI

```
I want to build a SaaS platform for managing online payments
for small businesses. The system needs to be scalable,
secure, and API-first.
```

Build Team

Team Builder using AI

📝 Project Analysis


```
{
  "project_type": "SaaS Platform",
  "domains": [
    "online payments"
  ],
  "required_roles": [
    {
      "role": "Software Architect",
      "seniority": "Senior",
      "skills": [
        "API design",
        "security"
      ]
    },
    {
      "role": "DevOps Engineer",
      "seniority": "Mid-level",
      "skills": [
        "cloud infrastructure",
        "containerization"
      ]
    }
  ],
  "tech_stack": [
    "Java",
    "Spring Boot",
    "AWS"
  ],
  "team_size": 5
}
```


🤖 Team Members

Sofia Rossi (API Engineer)

Seniority: Mid

Reason: Strong API design skills and experience with Spring Boot

Leila Haddad (Application Security Engineer)

Seniority: Senior

Reason: Proven track record of securing complex applications and expertise in security best practices

📌 Coordinator Report

Based on the provided project analysis and selected team, I'll provide my assessment:

Why each member was selected:

Sofia Rossi (backend_005) was chosen for her strong API design skills and experience with Spring Boot, which aligns well with the required role of Software Architect. Her mid-level seniority also fits within the project's requirements.

Leila Haddad (security_006) was selected due to her proven track record in securing complex applications and expertise in security best practices, making her an excellent fit for the Application Security Engineer role.

Team strengths:

The team has a strong foundation in API design and security, which are critical components of a SaaS platform. Sofia's experience with Spring Boot will help with integrating the tech stack, while Leila's expertise in application security ensures that the project is well-protected.

Weaknesses or risks:

One potential risk is the lack of DevOps Engineer representation. Although the team has some cloud infrastructure and containerization skills, a dedicated DevOps Engineer would provide more comprehensive support for deployment and maintenance.

Another concern is the imbalance in seniority levels. With only one Senior-level member (Leila), there may be a gap in leadership and decision-making abilities. This could lead to slower decision-making or potential bottlenecks in the project's progress.

Missing roles or seniority imbalance:

To address these concerns, I recommend adding a DevOps Engineer with mid-level seniority to ensure comprehensive support for deployment and maintenance. Additionally, considering a second Senior-level member (e.g., Software Architect) would help balance out the team's leadership and decision-making capabilities.

Actionable recommendations:

1. Prioritize finding a DevOps Engineer with mid-level seniority to fill the gap.
2. Consider adding a second Senior-level member to balance out the team's leadership and decision-making abilities.
3. Leverage Sofia's experience with Spring Boot to optimize the tech stack and API design.
4. Utilize Leila's expertise in application security to ensure robust protection for the SaaS platform.

By addressing these concerns, the team can better align with the project requirements and mitigate potential risks, ultimately leading to a more successful outcome.

Mistral LLM : 9 Min
Embedding Model : Nomic

Team Builder using AI

I want to build a SaaS platform for managing online payments
for small businesses. The system needs to be scalable,
secure, and API-first.

Build Team

Team Builder using AI

📝 Project Analysis


```
{
  "project_type": "SaaS Platform",
  "domains": [
    "Online Payments",
    "Small Businesses"
  ],
  "required_roles": [
    {
      "role": "Software Architect",
      "seniority": "Senior",
      "skills": [
        "Scalability",
        "Security",
        "API Design"
      ]
    },
    {
      "role": "Full Stack Developer",
      "seniority": "Mid-Level",
      "skills": [
        "Node.js",
        "Express.js",
        "PostgreSQL",
        "REST API"
      ]
    },
    {
      "role": "Security Engineer",
      "seniority": "Senior",
      "skills": [
        "Cryptography",
        "Penetration Testing",
        "OWASP Top Ten"
      ]
    }
  ],
  "tech_stack": [
    "Node.js",
    "Express.js",
    "PostgreSQL",
    "REST API"
  ],
  "team_size": 5
}
```

🤖 Team Members

Ahmed Hassan (Software Architect)

Seniority: Senior

Reason: Strong background in scalable, secure API design and 11 years of experience.

Isabella Rossi (Full Stack Developer)

Seniority: Mid

Reason: 6 years of experience, proficient in required tech stack and available full-time.

Luis Fernandez (Security Engineer)

Seniority: Mid

Reason: 5 years of experience, proficient in security-related skills and available freelance.

📌 Coordinator Report

```
 1. Ahmed Hassan was selected as the Software Architect due to his strong background in scalable, secure
API design and 11 years of experience, making him well-suited to lead the technical direction of the
project. Isabella Rossi, a mid-level Full Stack Developer with 6 years of experience, was chosen for her
proficiency in the required tech stack and availability for full-time work. Luis Fernandez, despite being a
mid-level Security Engineer, was selected due to his expertise in cryptography, penetration testing, and
OWASP Top Ten, ensuring the platform's security.

2. The team boasts a mix of diverse skills, with Ahmed's focus on scalable architecture and API design,
Isabella's proficiency in full-stack development, and Luis's emphasis on security. This combination should
help create a robust and secure SaaS platform for small businesses.

3. Potential weaknesses or risks include the mid-level seniority of both Isabella and Luis, which may limit
their ability to handle complex issues that might arise during the project. Additionally, having only one
senior-level role (Ahmed) might lead to an imbalance in decision-making and overall project leadership.

4. While there are no missing roles at this time, it is essential to keep in mind that as the project
progresses, additional roles such as a DevOps Engineer or UX/UI Designer may become necessary. Moreover,
having more senior-level personnel could help address complex challenges and ensure smooth decision-making
processes.

5. To mitigate identified weaknesses and risks:
   - Consider bringing in a senior-level Full Stack Developer or Security Engineer to provide guidance and
support to the mid-level team members.
   - Establish clear roles and responsibilities for each team member to ensure an even distribution of
decision-making power and maintain a balanced team structure.
   - Regularly review project progress and adjust team composition if necessary, ensuring that the team has
the right mix of skills and experience to meet evolving project needs.
```

Llama3 vs Mistral

Evaluation :

- **Mistral is Better than Llama3 in Project Analysis**
- **The style of writing the explanation in the final step is clearly different between Llama3 & Mistral**

Future Directions

- Filters
- Team Matching using Rule Based